

EDUCATION

- **Sharif University of Technology, Tehran, Iran** GPA: 19.16/20 (4.00/4.00)
M.Sc. in Physics Sep 2024 – Present
 - **Selected Graduate Courses:** Special Relativity, Quantum Mechanics III, Electromagnetics III, Statistical Mechanics III, Quantum Field Theory 1, String Theory I, String Theory II, M-Theory, Algebraic Quantum Field Theory and Quantum Gravity and Black Holes. (T. Hartman)¹
- **Shahid Beheshti University, Tehran, Iran** GPA: 16.56/20(Physics) (3.31/4.00)
B.Sc. in Physics Sep 2019 – Sep 2024
 - **Selected Undergraduate Courses:** Quantum Field Theory 0, Differential Geometry, Solid State Physics, Mathematical Physics III, Linear Algebra 0 and Linear Algebra and Matrix
- **Atomic Energy High School, Tehran, Iran** GPA: 20/20 (4.00/4.00)
Diploma Sep 2016 – Sep 2019

RESEARCH INTERESTS

Quantum Gravity, String Theory, Algebraic Quantum Field Theory, Number Theory

HONORS AND AWARDS

- **The University Students National Physics Olympiad, IPhO** Iran
Ranked 2024
- **The National Master Studies Entrance Exam** Iran
Ranked 2024
- **The University Students National Physics Olympiad, IPhO** Iran
Ranked 2023
- **The National Master Studies Entrance Exam** Iran
Ranked 2023
- **World Mathematics Team Competition, WMTC** Iran
Gold Medal 2014

RESEARCH EXPERIENCES

- **Review of Algebraic Quantum Field Theory** 2026 - 2026
Under the supervision of Dr. Amin Faraji Astaneh at Sharif University of Technology, Tehran, Iran
 - We gave an overview of entanglement, von Neumann algebra, and the emergence of spacetime.
- **The Imaginary Phase Problem in the Gravitational Path Integral** 2025 - 2026
Under the supervision of Dr. Amin Faraji Astaneh at Sharif University of Technology, Tehran, Iran
 - In the gravitational path integral, the saddle point approximation introduces Polchinski's phase, which confuses state counting. In de Sitter gravity, an observer's perspective yields a von Neumann algebra type II, hinting at a connection between Polchinski's phase and gravitational state counting.
- **A generalization of the Selberg zeta function for flat space cosmologies** 2024 - Present
Under the supervision of Dr. Reza Fareghbal at Shaid Beheshti University, Tehran, Iran

¹Self Study

- We generalize the Selberg zeta function to flat space cosmologies, linking number theory and quantum gravity beyond AdS. Using conjectural properties of the Selberg zeta function, our approach provides a rigorous number-theoretic foundation that reproduces these results without assumptions.

- **Group theory and irreducible representations of the Poincaré group** 2023 - 2024
Under the supervision of **Dr. Siamak Sadat Gousheh** at **Shahid Beheshti University, Tehran, Iran**
 - We started from the basics of group theory and obtained their irreducible representation for Poincaré groups.
- **Supersymmetry in Quantum Mechanics by Generalized Uncertainty Principle** 2023 - 2024
Under the supervision of **Dr. Siamak Sadat Gousheh** at **Shahid Beheshti University, Tehran, Iran**
 - We investigated supersymmetry in quantum mechanics with the generalized uncertainty principle.
- **Supersymmetry in Quantum Mechanics for Dirac equation in FRW space-time** 2022 - 2023
Under the supervision of **Dr. Siamak Sadat Gousheh** at **Shahid Beheshti University, Tehran, Iran**
 - We constructed supersymmetry quantum mechanics in curved space-time and investigated what conditions can preserve this symmetry in FRW space-time.

TA EXPERIENCES

- **Electromagnetics III:** Dr. Mehdi Kargarian ('26)
- **Physics Laboratory I:** Dr. Shant Baghran ('24)
- **Mathematical Physics I:** Dr. Mohammad Aliakbari ('23)
- **Quantum Mechanics I:** Dr. Ali Sadeghi ('22)
- **General Physics I:** Dr. Siamak Sadat Gousheh ('22)
- **Electromagnetics II:** Dr. Ali Sadeghi ('21) , (As a corrector in mid-term and final exams.)
- **Mathematical Physics II:** Dr. Seyed Ali Hosseiny Esfidavajani ('21)
- **Analytical Mechanics II:** Dr. Mohammad Aliakbari ('21)
- **Mathematical Physics I:** Dr. Seyed Ali Hosseiny Esfidavajani ('20)
- **General Physics III:** Dr. Siamak Sadat Gousheh ('19)

SCHOOLS

- **4th Advanced School on Holography and Quantum Information Topics**, IPM, Tehran, Iran

SKILLS

- **Languages:** Python, C/C++, HTML/CSS, Mathematica, L^AT_EX
- **Soft Skills:** Problem Solving, Effective Communication, Teamworking

LANGUAGES

- **Persian:** Native
- **English:** Upper Intermediate